

Recycled Aggregates in Structural Concrete: CO₂ Curing Treatments for Recycled Concrete Aggregate

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EGH404 Research in Engineering Practice – Assessment 1B

Introduction

Construction and demolition waste generates over 12.7 billion tonnes annually worldwide, with over 70% being concrete, even though concrete requires naturally mined aggregate (Peiris et al., 2025). RCA provides a solution for both these problems, but since the porous nature of the mortar coating each grain increases water absorption and decreases aggregate quality, RAC can only be used for non-structural applications. CO₂ curing improves RCA, as well as mineralising the CO₂ in the mortar coating, improving the quality of aggregate and reducing CO₂ emissions (Gao et al., 2024).

The current paper reviews CO₂ curing and the research gap caused by conflicting results. It includes introduction, research problem and questions, thematic literature review, and conclusion.

Scope includes peer-reviewed journal articles published after 2016 obtained by searching for the terms “recycled concrete aggregate,” “accelerated carbonation,” “CO₂ curing,” and “co-treatment agent” on Scopus and Engineering Village databases. There are three Q1 articles forming the core and five additional pieces of literature, resulting in the total of eight sources. All literature considered belongs to the period between 2022 and 2026 since the methods of carbonation performed in a chamber are mostly published during this time period. There is no geographical constraint since the research covers the Middle East, South and East Asia, Europe, and Australia, and the mechanisms used are universal.

Research Problems and Question

CO₂ treatment can be an effective way to improve RCA, as CO₂ is absorbed rather than emitted (Gao et al., 2024). Nonetheless, different research works give conflicting results on the efficiency of CO₂ treatment. According to Shehadah et al. (2026), CO₂ treatment is not very effective; nevertheless, the studies of Benahmad et al. (2026) and Liu et al. (2026) show the high effectiveness of CO₂ treatment. Both of the latter include a CO₂-only test, but they disagree markedly on how much the added agent contributes, and each examines only one level of carbonation intensity.

This review is guided by a single research question: once CO₂ concentration during curing is properly controlled, does adding a co-treatment agent, such as alkaline slurry or biochar, still

produce a meaningful improvement in recycled aggregate concrete, or does controlled CO₂ curing alone achieve comparable results? Answering this matters because a co-treatment agent adds cost, processing time and chemical use, none of which is justified unless it delivers a benefit controlled carbonation cannot.

Review of Existing Literature

All three articles describe the same basic mechanism: carbonation causes calcium carbonate to precipitate within the pores of the mortar surrounding the aggregate. This densifies the interfacial transition zone (ITZ) and reduces water absorption. The articles are compared using three main themes, with any contradictions discussed and resolved in the final subsection. Additional information is provided in Appendix A.

Theme 1: Mechanical Properties

The compressive strength was tested in all three articles, with very different outcomes. As is clear from Row 1 of Appendix B, Shehadah et al. (2026) conducted six treatments on 28 concrete mixtures and found that dry-ice CO₂ treatment led to only a 3% increase in compressive strength, resulting in a conclusion that dry-ice CO₂ and acetic acid “generally did not enhance the properties of the aggregates or the concrete cast with them” (p. 17). Drum mixing was the best of all the treatments, leading to a 15% increase in compressive strength and 60% in water absorption (p. 16). The merit of drum mixing is that it is quite comprehensive.

On the contrary, Benahmad et al. (2026) segregated both variables into one mix design, which is a great advantage of this experiment. The treatment of specimens with chamber-controlled CO₂ curing led to the increase in compressive strength by 6.0% at 28 days relative to untreated RAC (p. 9), while under the same conditions the treatment with by alkaline slurry increased it by 11.0-13.3%, closer to the control level; the authors state that “the treatment with highly alkaline slurry and p curing had superior performance than only CO₂ curing treatment” (p. 10). Nevertheless, elastic modulus was still lower than in the case of control. Liu et al. (2026) support this statement in smaller proportions: under the same CO₂ curing conditions, 4% biochar led to 3.7 MPa at 28 days, which is approximately 8% (p. 8).

One of the limitations of all three experiments is the use of just one type of aggregate; what’s more, Shehadah et al. (2026) have used just one water-cement ratio.

Theme 2: Treatment Conditions

These differences in outcomes are attributed to the way exposure to CO₂ was controlled. Shehadah et al. (2026, p. 4) note their 24-hour dry ice exposure method lacked “control over CO₂ pressure/concentration, humidity, and temperature,” which is why this method proved to be less efficient. Both Benahmad et al. (2026) and Liu et al. (2026) used the sealed, controlled chamber method of curing.

Studies supporting this finding demonstrate that exposure control, but not exposure time, is the key. Jamil et al. (2025) carbonated RCA in a controlled chamber at 1 bar with controlled

temperature and humidity levels and were able to achieve increases in compressive strength of 16% and 18% for 100% treated RCA, which is five times higher than the result reported by Shehadah et al. (2026) obtained in two to three hours as opposed to 24.

Theme 3: Durability Performance

However, durability is important for structural certification, even though its quantification was inconsistent. No durability or freeze-thaw testing by Shehadah et al. (2026) is done, and hence, their conclusion of superior mechanical performance compared to CO₂ is based solely on strength.

However, Benahmad et al. (2026) found the shrinkage of concrete decreased by 25.9% (p. 11) and the creep of concrete decreased by 22.4% (p. 13), as well as investigated chloride penetration. In their study, Liu et al. (2026) exposed specimens to 300 cycles of sulfate exposure and freeze–thawing and forecasted that the life span would be prolonged from 2,232–3,192 days to 6,480–9,288 days (pp. 20–21) due to some process involving nanoindentation and NMR analysis. Nonetheless, these are Weibull extrapolations based on accelerated cycles. As the authors noted, their four-mix correlation “was not intended to develop a general strength prediction model for external use” (p. 13). Workability is decreased by 36% in high doses (p. 8).

However, in both cases, the outcomes are compared to the effect of CO₂ curing done independently. More specifically, the shrinkage when done with CO₂ alone was reduced by 17.7%, while using slurry resulted in a decrease by 25.9% (Benahmad et al., 2026, p. 11). In a similar way, Liu et al. (2026, p. 13) showed the proportion of harmful pores dropped from 50.44% to 18.38% in the presence of biochar under the same conditions.

Comparative Analysis and Contradiction Resolution

The most obvious contradiction is that Shehadah et al. (2026) demonstrate a marginal effect of CO₂ conditioning since the compressive strength increases merely by 3% (p. 17), whereas Benahmad et al. (2026) achieve strength comparable to natural aggregate (p. 10), and Liu et al. (2026) note that there was an enhancement of ITZ modulus from 17.68 to 24.87 GPa due to a combination of biochar and CO₂ conditioning (pp. 12-13). Hence, the effects differ significantly in terms of size.

The contradiction is of the methodological kind. Uncontrolled dry-ice exposure of Shehadah et al. (2026, p. 4) that was quoted above should be contrasted with controlled exposure in chambers used by the other two teams. The sample allocation makes the contradiction even more obvious since Shehadah et al. (2026) have allocated 28 mixes to six competing treatments (p. 5), having few specimens for each treatment, while Benahmad et al. (2026) used five mixes for one treatment at four ages (p. 4) and Liu et al. (2026) allocated ten mix-curing combinations (p. 4).

Addressing this issue highlights another problem: both positive studies isolate variables, yet differ in their results. According to Benahmad et al. (2026, p. 10), the agent almost doubles the benefits of CO₂ treatment from 6.0% to 11.0–13.3%, while Liu et al. (2026, p. 8) state that the agent raises this figure by only 8%. Carbonation alone is unstable and not consistent in terms of intensity, e.g. $20 \pm 3\%$ CO₂ concentration at atmospheric pressure results in 6.0% (Benahmad et al., 2026, pp. 3, 9), 90% CO₂ at 0.15 MPa results in 5.7% (Liu et al., 2026, pp. 5, 8), whereas 1 bar CO₂ for 2-3 hours results in 16-18% (Jamil et al., 2025). Evidently, concrete grade and initial quality of aggregates play a big role, as much as the intensity does.

The literature does not address the problem. Jamil et al. (2025) perform controlled carbonation of concrete yet do not have the agent arm, hence they cannot estimate the benefits of the agent; Al-Waked et al. (2022) provide a comparison between the carbonation with and without limewater agent, yet use only aggregate-level indicators. And Kosuri et al. (2023) optimise impregnation of already-carbonated aggregate with an agent, yet not comparing it with non-impregnated aggregate. No literature considers varying the intensity of carbonation and presence of the agent together, hence there is no clarity whether the agent provides a mechanism unavailable for carbonation or is just a compensation for weak carbonation. Moreover, no study compares CO₂ arm alone across both strength and durability. This is the research gap highlighted by this review.

Conclusion

This review has shown that CO₂ curing may considerably enhance recycled aggregate concrete if the exposure conditions are controlled. The contradiction between Shehadah et al. (2026) and the beneficial results of Benahmad et al. (2026) and Liu et al. (2026) is due to the lack of concentration, humidity and temperature control in the former study. What is more important, the two studies that isolate the agent differ in its effect – almost twice in the first case and an extra 8% in the second –, whereas the carbonation-only improvement varies between 6% and 18% without considering the exposure intensity in all the considered literature sources.

Practically speaking, there is still debate about the necessity and justification of the energy used to create an alkaline slurry or biochar or the chemicals themselves, or whether just CO₂ curing is sufficient. Application of a needless reagent represents poor resource utilization in a process that is inherently high-emitting, while the omission of a critical one may risk the longevity of public works projects. Reporting the impact of each reagent individually therefore represents a professional responsibility, especially considering that the lifetimes of the material calculated here have been based on accelerated tests rather than actual usage data. Further studies should vary both carbonation level and reagent application in the same aggregate/mix design, using a natural aggregate baseline for comparison.

References

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Appendix A: CRAAP Analysis

Table 1: CRAAP Evaluation of all sources used.

Paper (Author, Year, Journal)	Currency	Relevance	Authority	Accuracy	Purpose
Shehadah et al. (2026), Case Studies in Construction Materials	Very current — 2026	Compares CO ₂ with five RCA treatments	Q1 Scimago; Elsevier; KFUPM	28 mixes; multiple testing methods; limitations stated	Compares RCA treatment effectiveness
Benahmad et al. (2026), Results in Engineering	Very current — 2026	Tests CO ₂ curing with alkaline slurry	Q1 Scimago; Elsevier; KMUTT	Tested from 3–90 days across key properties	Develops high-strength RAC
Liu et al. (2026), Case Studies in Construction Materials	Very current — 2026	Explains combined treatment mechanisms	Q1 Scimago; Elsevier; international authors	Nanoindentation and NMR; limitations stated	Examines RAC durability and microstructure
Jamil et al. (2025), Archives of Civil and Mechanical Engineering	Current — 2025	Examines carbonation pressure and duration	Q1 Scimago; Springer; Scopus	Controlled conditions; multiple verification methods	Optimises RCA carbonation
Al-Waked et al. (2022), Case Studies in Construction Materials	Current — 2022	Compares carbonation with/without limewater	Q1 Scimago; Elsevier; USW	Tests CO ₂ levels, time and coating thickness	Evaluates RCA treatment efficiency
Gao et al. (2024), Cement and Concrete Research	Current — 2024	Compares carbonation with/without sodium bicarbonate	Q1 Scimago; Elsevier; PolyU	Experiments supported by simulations	Improves CO ₂ treatment efficiency
Kosuri et al. (2023), Journal of Materials in Civil Engineering	Current — 2023	Optimises slurry treatment of carbonated RCA	Q1 Scimago; ASCE; IIT Madras	Tests treatment variables and paving concrete	Supports large-scale RCA use
Peiris et al. (2025), Environmental Science and Pollution Research	Very current — 2025	Reviews 150+ RCA treatment studies	Q2 Scimago; Springer; RMIT	Systematic review with selection criteria	Summarises RCA treatment research

Appendix B: Triangulated Summary Table

Table 2: Triangulated summary of primary papers

Paper (Author, Year, Journal Name)	Design	Sample Size	Key Measures	Reported Effect Size / Stats	Page Number(s)	Main Findings	Reported Limitations	Other Identified Limitations	Notes on Contradictions
Shehadah et al. (2026), Case Studies in Construction Materials	Lab experiment comparing six RCA treatments with natural and untreated controls	28 mixes; n=3 per test (n=1 for LA abrasion)	Density, absorption, abrasion, strength, UPV, SEM, XRD	Mechanical mixing: compressive +15%, tensile +16%, absorption -60%; CO ₂ : +3% compressive strength	Design p.4; Sample p.5; Measures p.6; Effect p.16; Findings p.17; Limits p.4	Mechanical mixing was most effective; dry-ice CO ₂ was generally ineffective	CO ₂ exposure lacked control of pressure, humidity and temperature	One RCA source; one w/c ratio; no durability or freeze-thaw testing	Contradicts Benahmad and Liu, but used uncontrolled CO ₂ exposure
Benahmad et al. (2026), Results in Engineering	Lab experiment comparing five mixes, including CO ₂ -only and slurry + CO ₂ treatments	5 mixes; n=3 per test; tested at 3, 7, 28 and 90 days	Density, absorption, abrasion, strength, modulus, shrinkage, creep, chloride migration	Slurry + CO ₂ : near-control strength; shrinkage -25.9%; creep -22.4%	Design pp.3-4; Sample p.4; Measures pp.4-6; Shrinkage p.11; Creep p.13; Findings p.14	Controlled CO ₂ curing with alkaline slurry achieved near-natural strength and improved durability	No formal limitations section; modulus remained below the control	One lab-crushed RCA source; fixed slurry conditions; testing limited to 90 days	Opposes Shehadah; shows CO ₂ can work under controlled conditions with an agent
Liu et al. (2026), Case Studies in Construction Materials	Lab experiment using five biochar levels under normal and CO ₂ curing, followed by 300 freeze-thaw cycles	5 × 2 combinations; n=3 per test; correlation n=4	Strength, carbonation depth, ITZ modulus, pore structure, predicted service life	4% biochar + CO ₂ : ITZ 17.68 → 24.87 GPa; harmful pores 83.2% → 18.4%	Design p.3; Sample p.3; Measures pp.6-8; ITZ/pores pp.12-13; Service life pp.20-21; Limits p.13	Controlled CO ₂ + 4% biochar improved ITZ, pore structure and predicted service life	Correlation analysis (n=4) not intended as a general prediction model	Weibull-based service-life prediction; one biochar source; workability reduced up to 36%	Supports Benahmad: controlled CO ₂ with an agent can improve performance